Min Jae Bae

Ph.D. Candidate · Plant Stress Epigenetics and Chromatin Biology

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SUMMARY

Plant molecular biologist studying how chromatin regulators shape cold stress responses in *Arabidopsis*. My PhD links the $CRL4^{HOS15}$ E3 ligase to the turnover of linker histone H1.3 and the histone deacetylase HD2C, using genetics, biochemistry, chromatin accessibility (NicE-seq), RNA-seq and LC-MS proteomics, with all sequencing analysis done in R. Member of the NASA GeneLab Plant Analysis Working Group. PhD defense November 2026; degree conferred February 2027; available for a postdoctoral position from March 2027.

EDUCATION

Ph.D., Biomedical Science and Engineering, Konkuk University, Seoul — Mar. 2021 – Feb. 2027 (expected)

Advisor: Prof. Dae-Jin Yun. Dissertation on HOS15-dependent turnover of linker histone H1.3 and LisH-mediated HOS15 self-association in the cold response. Defense: November 2026.

M.S., Biomedical Science and Engineering, Konkuk University, Seoul — Mar. 2019 – Feb. 2021

Thesis: Structural and Functional Analysis of HOS15, a Substrate Receptor of CRL4 in Cold Stress Signaling. Advisor: Prof. Dae-Jin Yun.

B.S., Biomedical Science and Engineering, Konkuk University, Seoul — Mar. 2013 – Feb. 2019

Thesis: Functional Studies of SUMOylation of Histone Deacetylase in Plants. Studies interrupted for military service, 2014–2015.

RESEARCH EXPERIENCE

Doctoral research, Plant Stress Epigenetics, Konkuk University (Prof. Dae-Jin Yun) — 2021 – present

Linker histone H1.3 in the cold response

- Followed up linker histone H1, previously found among HOS15-associated proteins by IP-MS, and confirmed the HOS15–H1.3 interaction in yeast and *N. benthamiana*.
- Showed that cold lowers H1.3 protein in a HOS15-dependent, MG132-sensitive manner (immunoblot and native-promoter H1-GFP lines).
- Found that *h1.3* mutants have higher basal (non-acclimated) freezing tolerance and maintain rosette growth at 4 °C.
- Integrated NicE-seq, RNA-seq and LC-MS data: *h1.3* gains accessibility mainly in pericentromeric regions, and before cold already expresses part of the wild-type cold-induced stress and phenylpropanoid program, while CBF-target COR genes are not pre-induced.

HOS15 self-association and HD2C degradation

- Showed that HOS15 self-associates through its N-terminal LisH motif; alanine substitutions of conserved hydrophobic LisH residues abolish self-association in yeast two-hybrid assays.
- Generated complementation lines showing that these LisH variants fail to support cold-induced ubiquitination and degradation of HD2C and do not restore the freezing tolerance of *hos15-2*.
- Predicted disordered regions of HOS15 from structure models (AlphaFold-type models and IDR predictors).

Spaceflight omics re-analysis, NASA GeneLab Plant AWG — Sep. 2026 – present

- Began re-analysis of public plant spaceflight and radiation RNA-seq datasets in NASA's Open Science Data Repository (OSDR) to test terrestrial stress gene signatures.

PUBLICATIONS

1. **Bae MJ**, Zareen S, Yun H, Ali A, Bader ZE, Park K, Albakri N, Jung YJ, Yun DJ, Park J. Unveiling the epigenetic and stress-responsive function of histone H1 in plants. *Plant Signaling & Behavior* (2026). doi:10.1080/15592324.2026.2677313
2. Ali A, Zareen S, Bader ZE, Park J, Khan IU, Park K, Albakri N, **Bae MJ**, Bressan RA, Pardo JM, Yun DJ, Xu ZY. A GPS2-like protein interacts with HOS15 and HDA6 to form a repressor complex that regulates ABA signaling and drought adaptation in Arabidopsis. *Plant Communications* (2026). doi:10.1016/j.xplc.2026.101843
3. Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. Arabidopsis E3 ligase HOS15 regulates ABA and drought responses via DIL9 degradation. *Journal of Plant Growth Regulation* (2026). doi:10.1007/s00344-026-12240-w
4. Albakri N, Bader ZE, Park KS, **Bae MJ**, Zareen S, Yun H, Pradhan S, Yun DJ, Park J. HOS15 modulates alternative splicing homeostasis under cold stress in *Arabidopsis thaliana*. *Journal of Plant Biology* (2025). doi:10.1007/s12374-025-09465-9
5. Park K, Lim CJ, Albakri N, **Bae MJ**, Park J. HOS15 contributes to thermotolerance through destabilization of HD2C in Arabidopsis. *Journal of Plant Biology* (2025). doi:10.1007/s12374-025-09466-8
6. Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. HOS15 together with PWR–HDA9 positively regulates dark-induced senescence in Arabidopsis. *Plant Signaling & Behavior* (2025). doi:10.1080/15592324.2025.2564962
7. Bader ZE[†], **Bae MJ**[†], Ali A, Park J, Baek D, Yun DJ. GIGANTEA-ENHANCED EM LEVEL complex initiates drought escape response via dual function of ABA synthesis and flowering promotion. *Plant Signaling & Behavior* (2023). doi:10.1080/15592324.2023.2180056
8. Lim CJ, Park KS, Ali A, Park J, Ryou SM, Shen M, Khan HA, Bader ZE, Zareen S, **Bae MJ**, Choi JH, Xu ZY, Pardo JM, Yun DJ. Negative regulation of floral transition in Arabidopsis by HOS15-PWR-HDA9 complex. *Frontiers in Plant Science* (2023). doi:10.3389/fpls.2022.1105988
9. Lim CJ, Park J, Shen M, Park HJ, Cheong MS, Park K, Baek D, **Bae MJ**, Ali A, Jan M, Lee SY, Lee BH, Kim WY, Pardo JM, Yun DJ. The histone-modifying complex PWR/HOS15/HD2C epigenetically regulates cold tolerance. *Plant Physiology* (2020). doi:10.1104/pp.20.00439

[†] These authors contributed equally.

Manuscripts in preparation

- **Bae MJ**, Bader ZE, Ali A, Yun DJ, Park J. Mechanisms of histone H1 degradation under cold stress.
- Ali A[†], **Bae MJ**[†], Bader ZE, Zareen S, Park B, Ali Z, Yun DJ, Park J. HOS15 dimerization promotes HD2C degradation during cold stress.

PATENT APPLICATIONS (KOREA; CO-INVENTOR)

- Vector for improving productivity of cellulose and transformed mouse-ear cress by the same. KR 10-2023-0113597
- Vector for improving productivity of cellulose and transformed mouse-ear cress by the same. KR 10-2024-0168432
- Vector for improving productivity and storage capacity of vegetable oil and transformed plant by the same. KR 10-2024-0168464
- Ethanol production and tolerance enhanced yeast strain and method for producing ethanol using the same. KR 10-2025-0104960
- Sucrose-accumulating plant and method for preparing the same. KR 10-2025-0104961
- Recombinant *Kluyveromyces marxianus* strain with engineered xylose metabolic pathway and method for producing ethanol using the same. KR 10-2025-0157597

SKILLS

- **Plant molecular biology and genetics:** Gateway cloning, Gibson assembly, site-directed mutagenesis, vector design (Vector NTI), transgenic and complementation lines, *Arabidopsis* genetics, freezing tolerance and stress phenotyping

- **Chromatin and protein biochemistry:** NicE-seq, chromatin immunoprecipitation and ChIP-seq sample preparation, nuclear fractionation, immunoblotting, cycloheximide chase and proteasome-inhibitor assays, in vivo ubiquitination assays, yeast two-hybrid, transient expression in *N. benthamiana*, confocal imaging
- **Omics data analysis:** RNA-seq (Salmon, DESeq2), gene set enrichment analysis (GSEA), GO and KEGG enrichment, NicE-seq peak and promoter accessibility analysis (HOMER, IGV), LC-MS proteomics data analysis, public data re-analysis (NASA OSDR, GEO)
- **Software development:** PhytoOmics Studio, an R/Shiny application for plant multi-omics analysis (in development; RNA-seq module with DESeq2, GO/KEGG enrichment and GSEA for four plant species; Windows installer with bundled R)
- **Software/tools:** R and Bioconductor, GraphPad Prism, Bio-Rad CFX Maestro (RT-qPCR), Zeiss ZEN and Leica LAS X (confocal imaging), ImageJ

RESEARCH PROJECTS

- **Global Plant Stress Research Center** (NRF), 2024.08–2030.04 — Participating researcher
- **Genome-wide chromatin reprogramming and transcriptome regulation in plant environmental stress adaptation** (NRF), 2026.03–2031.02 — Participating researcher
- **Chromatin reconfiguration and transcriptional control for plant stress adaptation** (NRF), 2022.03–2027.02 — Participating researcher

PROFESSIONAL ACTIVITIES

- Member, NASA GeneLab OSDR Plant Analysis Working Group (Plant AWG), Sep. 2026–present
- Member, NASA GeneLab OSDR Regolith Analysis Working Group (Regolith AWG), Sep. 2026–present
- Organizing committee, 2nd Global Plant Stress Research Center (GPSRC) & 2025 KSMCB Plant Molecular Biology International Symposium, Seoul, Aug. 2025

AWARDS AND HONORS

HOPE Fellow, 15th HOPE Meeting with Nobel Laureates, Japan Society for the Promotion of Science (JSPS), Kyoto — Feb.–Mar. 2024

Excellent Poster Award, Korean Society of Plant Biologists (KSPB) — Nov. 2021

PRESENTATIONS

- 2026 GPSRC International Conference, Konkuk University, Seoul, May 2026
- 2026 Winter Conference of the Korean Society of Plant Biologists, Korea University, Seoul, Feb. 2026 — Poster
- 2024 International Conference of the Korean Society of Plant Biologists, Seogwipo, Nov. 2024 — **Oral presentation**
- 15th HOPE Meeting with Nobel Laureates, Kyoto, Japan, Feb. 2024 — Poster
- International Conferences of the Korean Society of Plant Biologists, 2018, 2020, 2021, 2022; Annual Conference, 2019 — Posters

LANGUAGES

Korean (native); English (fluent; American international school, Shenzhen, China, 2005–2012); Mandarin Chinese (conversational)

MILITARY SERVICE

Republic of Korea Marine Corps, airborne unit; mandatory service completed — Mar. 2014 – Dec. 2015

REFERENCES

Available on request.